OXYBUTYNIN

Cognitive impairment in elderly

Dry mouth

CHEMICAL NAME 4-(diethylamino)but-2-ynyl (RS)-2-cyclohexyl-2-hydroxy-2-phenylacetate (hydrochloride)

CLASS Antimuscarinic bladder antispasmodic

TRADE NAMES Ditropan · Kentera (patch) · Lyrinel XL · Cystrin

THERAPEUTIC USES / INDICATIONS

- Overactive bladder: urgency, frequency, urge incontinence.
- Neurogenic detrusor overactivity (spina bifida, spinal injury).
- Nocturnal enuresis (children, with alarm); bladder spasm after surgery/catheter.

ADMINISTRATION

- **PO** IR 2–3 times daily or MR once daily (swallow whole).
- **Transdermal** patch twice weekly (fewer side effects).
- **Intravesical** instillation (specialist).

DOSAGES

- **Adult:** 5 mg 2–3 times daily (max 5 mg four times daily); MR 5–20 mg once daily.
- **Elderly/frail:** 2.5 mg twice daily initially (up to 5 mg twice daily).
- **Children ≥5 yr:** 2.5 mg twice daily, up to 5 mg 2–3 times daily.
- **Patch:** 3.9 mg/day applied twice weekly (every 3–4 days).

PHARMACOKINETICS

- **Onset:** 30–60 min. **Duration:** 6–10 h (MR 24 h).
- Extensive first-pass metabolism (CYP3A4) to active N-desethyl-oxybutynin (more side effects).
- Excreted in urine. **t½:** 2–3 h.

PHARMACODYNAMICS

- Blocks **muscarinic (M3) receptors** on detrusor → ↓ involuntary contractions, ↑ bladder capacity.
- Direct antispasmodic and local anaesthetic effects.
- Crosses blood–brain barrier.

SIDE & ADVERSE EFFECTS

SIDE EFFECTS (COMMON)

- Dry mouth (very common), constipation.
- Blurred vision, dry eyes.
- Drowsiness, dizziness.

ADVERSE EFFECTS (SERIOUS)

- Cognitive impairment, confusion, delirium (elderly); avoid in dementia.
- Urinary retention.
- Heat stroke (reduced sweating).
- Angle-closure glaucoma; tachycardia.

CONTRAINDICATIONS & CAUTIONS

- Urinary retention, significant bladder outflow obstruction.
- Angle-closure glaucoma; myasthenia gravis.
- GI obstruction, severe ulcerative colitis, toxic megacolon.
- Frail elderly / dementia (relative).

KEY INTERACTIONS

- **Other anticholinergics:** additive burden.
- **Donepezil, rivastigmine:** opposing effects.
- **CYP3A4 inhibitors:** ↑ levels.

NURSING MANAGEMENT

- Measure post-void residual if retention suspected.
- Monitor cognition in older adults.
- Bladder diary; bladder training.
- Bowel and oral care.

PATIENT EDUCATION

- Sip water/sugar-free gum for dry mouth.
- Avoid overheating; drink fluids.
- Report confusion, difficulty passing urine, eye pain.
- Combine with bladder training and pelvic floor exercises.

KEY POINTS

- Antimuscarinic for overactive bladder.
- Dry mouth, constipation.
- 5 mg BD–TDS.
- Confusion in elderly — prefer patch or alternatives.

SOURCES CHECKED **[699]** Ditropan 5mg Tablets (UK SmPC) **[700]** Oxybutynin hydrochloride 10 mg prolonged-release tablets (UK SmPC) **[701]** Oxytrol, patch (US label) **[702]** oxybutynin, oral (US label) **[703]** oxybutynin, oral (US label, NCS HealthCare of KY)

Full references p. 432–457